Quiz 1 - DSC 10, Winter 2025

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

Instructions:

- This quiz consists of 4 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

The DataFrame `clubs` contains information about **all** the studemt organizations at UCSD. For each club, we have its `"Name"`, `"Category"`, number of `"Members"`, `"Budget"` in dollars, `"Rating"` out of 5.0, and whether it is `"Active"` represented by either `True` or `False`. The first three rows of `clubs` are shown below. Assume that all club names are unique.

	Name	Category	Members	Budget	Rating	Active
0	Cheese Club	Social	50	800	4.6	True
1	Data Science Student Society	Pre-Professional	300	500	5.0	True
2	Painting Club	Artistic	84	0	3.7	False

Question 1

The Cheese Club currently has 50 members. Each quarter, the number of people who will join the club is the value of the Python expression below.

```
3 * ((4 * 20 / 5 * 2 + 4) ** 0.5 - 2 ** 2)
```

How many members will the club have **after 3 quarters**?

◯ 56 ◯ 68 ◯ 74 ◯ 82 ◯ None of these.

Question 2

a) How many unique data types exist in the `clubs` DataFrame?

◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ None of these.

b) Suppose we run the following code.

```
clubs_and_ratings = ["Cheese Club".upper(), str(int(4.6)),
                     "Painting Club".title(), 3.7 - 1]
a = np.array(clubs_and_ratings)
```

What does each of the following expressions evaluate to?

`a[0]`: _____ `a[1]`: _____

`a[2]`: _____ `a[3]`: _____

PID: _____

Question 3

a) One of the clubs in the `clubs` DataFrame is the `"Surf Club"`. **Without** querying, write an expression that evaluates to the rating of this club.

b) Now, using a query, write an expression that evaluates to the rating of the `"Surf Club"`.

c) Fill in the blanks so the expression below evaluates to the name of the category (e.g. `"Social"`) with the largest number of members, on average.

```
(clubs.groupby(__(a)__).__(b)__
      .sort_values(by="Members", ascending=__(c)__).__(d)__)
```

`(a)`: `(b)`: `(c)`: `(d)`:

d) In each category, what is the largest budget for a club? Write one line of code that would produce an appropriate **data visualization** to answer this question.

Question 4

The Data Science Student Society has a `$500` budget, which they will use only to buy food for their 4 meetings this quarter. Each time they have a meeting, they will use exactly half of their remaining budget on food. For example, the first meeting's food cost will be `$250` and the next meeting's food cost will be `$125`. Use `np.arange()`, together with arithmetic operations, to assign `food_costs` to an array containing the food cost for each of the 4 meetings. **Do not** use `np.array()`.

`food_costs =`

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.